M359 Hospital conceptual data model

This conceptual data model has been developed from the Hospital scenario example in *Block 1*, Section 5. The model is also used to support the course material in *Blocks 2* and *3*.

The details are supplied here so that you can see a complete conceptual data model for the Hospital scenario.

This database card should be useful to you throughout the course.

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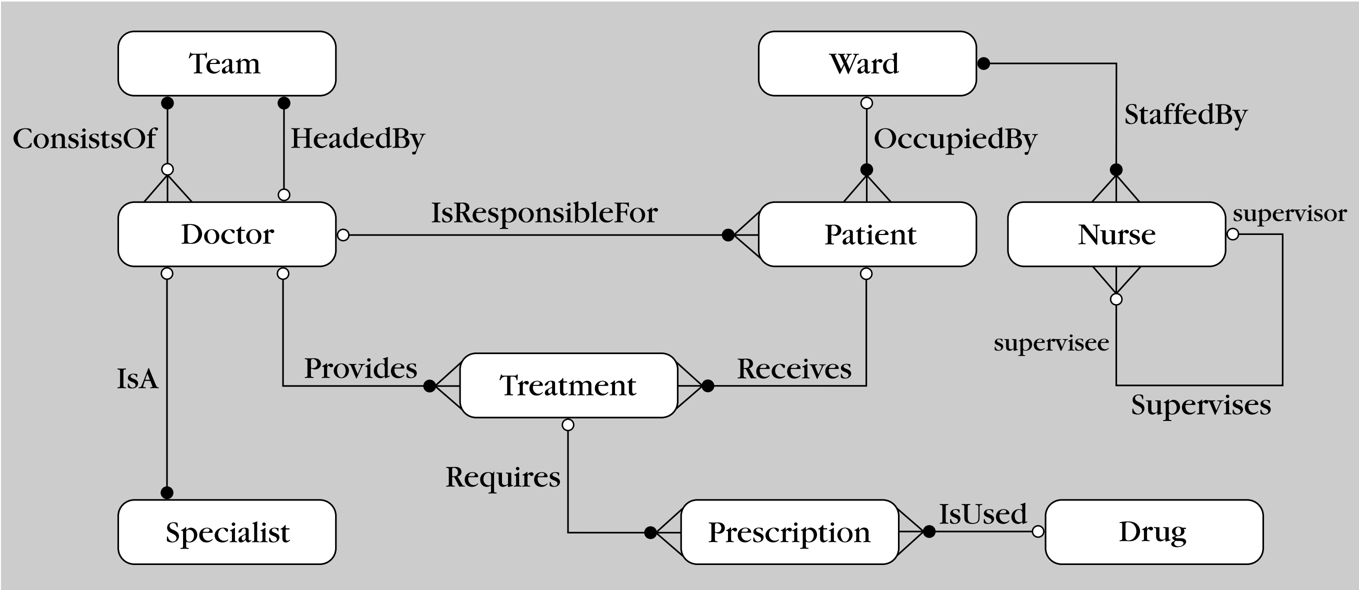
Entity–relationship diagram



Entity types

Ward (<u>WardNo</u>, WardName, NumberOfBeds)
Patient (<u>PatientId</u>, PatientName, Gender, Height, Weight)
Nurse (<u>StaffNo</u>, NurseName)
Doctor (<u>StaffNo</u>, DoctorName, Position)
Specialist (<u>StaffNo</u>, Specialism)
Team (<u>TeamCode</u>, TelephoneNumber)
Treatment (<u>StaffNo</u>, <u>PatientId</u>, <u>StartDate</u>, Reason)
Drug (<u>DrugCode</u>, DrugName, Type, Price)
Prescription (<u>PrescriptionNo</u>, Quantity, DailyDosage)

Additional constraints

c.1 Each doctor who is responsible for a patient must be a consultant. That is, an instance of the entity type Doctor that is involved in the relationship IsResponsibleFor must have a position of consultant.

c.2 A doctor who is the head of a team must be a consultant. That is, an instance of the entity type Doctor that is involved in the relationship HeadedBy must have a position of consultant.

c.3 A consultant must head a team. Doctors who are not consultants must be members of a team. That is, an instance of the entity type Doctor whose position is that of a consultant belongs to a team via the HeadedBy relationship, whereas other doctors belong to a team via the ConsistsOf relationship.

c.4 A doctor who treats a patient must be in the same team as the consultant who is responsible for that patient. That is, an instance of the entity type Doctor that provides treatment for a Patient must be from the same Team as the consultant who is responsible for that same Patient.

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c.5 A nurse can only supervise nurses in the same ward. That is, the two instances of the entity type Nurse that are involved in an occurrence of the Supervises relationship must be assigned to the same ward.

c.6 Doctors can be consultants, registrars or house officers. That is, the attribute Position (of entity type Doctor) may have a value of Consultant, Registrar or House Officer.

c.7 Only consultants can have specialisms. That is, an instance of the entity type Doctor can only be involved in the IsA relationship if the value of the Position attribute is Consultant.

c.8 Treatment is a weak entity type dependent on the entity type Doctor. So, each value of the StaffNo attribute in the entity type Treatment must be the same value as the StaffNo attribute of the Doctor instance to which the Treatment entity type is related by the relationship Provides (a consequence of weak–strong entity types).

c.9 Treatment is a weak entity type dependent on the entity type Patient. So, each value of the PatientId attribute in the entity type Treatment must be the same value as the PatientId attribute of the Patient instance to which the entity type Treatment is related by the relationship Receives (a consequence of weak–strong entity types).

c.10 Specialist is a weak entity type dependent on the entity type Doctor . So, each value of the StaffNo attribute in the entity type Specialist must be the same value as the StaffNo attribute of the Doctor instance to which the entity type Specialist is related by the relationship IsA (a consequence of weak–strong entity types).

c.11 The number of patients in a ward cannot exceed the number of beds in that ward.

c.12 A consultant must have a specialism. That is, an occurrence of the entity type Doctor that has the value Consultant for the Position attribute must be involved in an occurrence of the IsA relationship.

c.13 A nurse cannot have the same staff number as a doctor. That is, the value of the StaffNo attribute for each occurrence of the Nurse entity type cannot be the same as the value of the StaffNo attribute for any occurrence of the Doctor entity type.

Assumptions

None remaining.

Limitations

l.1 Only the details of a patient's current stay in hospital are recorded (i.e. only as an in-patient with no patient history of previous stays in the hospital).

l.2 Each prescription can only include details of a single drug.

l.3 Each patient must be allocated to a ward.